

WEEK 1 ASSIGNMENT AI CONCEPTS REPORT

Q1:What is Artificial Intelligence?

A: It is a branch of Computer Science and Engineering. It focuses on building machines that are capable of performing tasks that typically require human knowledge. Now-a-days it has been going to replace humans. In our everyday day life we use ai in many forms like chat gpt, perplexity, gemini etc.

Q2:Difference between AI, Machine Learning, Generative AI, and Agentic AI

A: AI- It focuses on logic & problem - solving

Machine Learning - It is a data driven method

Generative AI - It creates new content

Agentic AI - It acts through multiple steps of planning

Q3:Explain any three open-source AI models available on Hugging Face

A: Meta's Llama3 - It is developed by Meta.This is one of the most popular open-source Large Language Models (LLMs). It excels at general reasoning, coding, and creative writing tasks

Mistral 7B - It is released by Mistral AI.This model is highly efficient for coding. It uses advanced techniques like Sliding Window Attention etc

Stability AI - This model converts text into image. It is used to generate high quality images as well as realistic.

Q4: What is the difference between traditional programming and AI-assisted development?

A: Traditional Programming - It is a rule based process where it depends on humans to write codes using some instructions for computers to follow.

AI-assisted development - It uses machines and AI assistants for code generation and code debugging based on human prompts.

Q5: Explain AI hallucinations with one example

A: When an AI confidentially give false answers as true and present them as true but actually there are false

Ex:A fake biography - when we ask AI to write a biography of ourselves. It means we didn't give any information but it gives us our biography it means it gives false information and wanted to prove that it was correct.

AN AI GENERATED IMAGE

